



Module 4 Transformers BESCK204B

1 st sem cse (Visvesvaraya Technological University)



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MODULE 4

TRANSFORMERS

NECESSITY OF TRANSFORMER:

Transformer is a static device which transfers electric energy from one electric circuit to another at any desired voltage without any change in frequency. Transformers can increase or reduce the alternating current during generation, transmission, distribution and utilization of electric power. For instance, in transmission lines for high voltage generation, for stepping down the voltage in motors, lamps, heaters and other loads. Medium sized transformer has efficiency of 97-98%. As the transformer is static apparatus, there are no moving parts and maintenance is easy. They are even used in electronic circuits for low power applications.

PRINCIPLE OF OPERATION OF SINGLE PHASE TRANSFORMER:

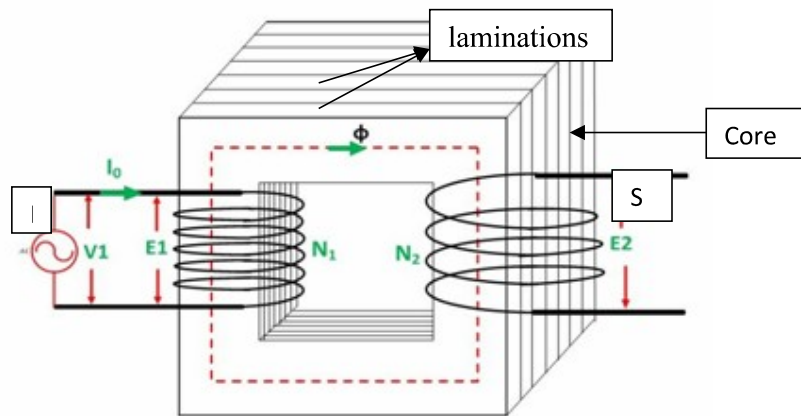


Fig 1: basic transformer

A transformer works on the principle of electromagnetic and mutual induction. “Whenever a change in current takes place in a coil there will be an induced emf in the other coil wound over the same magnetic core”. This is the principle of mutual induction by which the two coils are said to be coupled with each other.

Fig 1 shows the general arrangement of a **single phase transformer**. Core is made of **laminated steel sheets** of about 0.4 – 0.7mm thick insulated from one another by varnish or thin paper. The purpose of laminating the core is to reduce the power loss due to eddy currents induced by the alternating magnetic flux. The vertical portions of the core are called limbs and the top and bottom portions are called the yokes. Coils P and S are wound on the limbs. Coil P is connected to the supply and therefore called as the primary, coil S is connected to the load and is called as the secondary. An alternating voltage applied to P drives an alternating current through P and this current produces an alternating flux in the iron core, the mean path of the flux is represented by the dotted line ϕ . This flux links with the coil S and thereby induces an emf (E_2) in S.

Flux ϕ produces self induced emf E_1 in the primary winding P, and ϕ links with secondary to produce mutually induced emf E_2 in the secondary winding.

The RMS values of emf's are $E_1 = -N_1 \frac{d\phi}{dt}$ and $E_2 = -N_2 \frac{d\phi}{dt}$

Therefore, $\frac{E_1}{E_2} = K$, where K is the voltage transformation ratio.

The frequency of the two emf's is the same. The emf per turn is the same for P and S. When the secondary is open circuit, its terminal voltage is the same as the induced emf (E_2).

As power transferred from primary to secondary winding is same,

$V_1 I_1 \times \text{primary power factor} = V_2 I_2 \times \text{secondary power factor}$. As both the power factors are equal on full load, $\frac{V_1}{V_2} = \frac{I_2}{I_1} = K$.

Therefore, $\frac{E_1}{E_2} = K$

TYPES AND CONSTRUCTION OF SINGLE PHASE TRANSFORMERS:

There are two basic circuits in a transformer, magnetic circuit and electric circuit. The core forms the magnetic circuit and the electric circuit consists of two windings primary and secondary. The windings are insulated from each other and from the laminated steel core. The assembled core and the windings are enclosed in a suitable container. Porcelain or capacitor bushings are used to insulate and bring out the terminals of the windings from the enclosure.

Core is made up of laminations fitted together in such a way so as to ensure continuous magnetic path, with minimum air gap. Steel laminations has high silicon content, to provide high permeability and low hysteresis loss at the usual operating flux densities. Laminations minimize eddy current loss. They are insulated from each other by thin coating of varnish. Thickness of laminations ranges from 0.35mm to 0.5mm.

i) Core type transformers:

Core is surrounded by the windings. In the figure 1, primary and secondary are wound on the opposite limbs. But in practice, half the primary and half the secondary is situated side by side on each limb to reduce the leakage flux. For small transformers, a rectangular core is used with cylindrical coils. For large transformers, round or cylindrical coils are wound over a cruciform core section in helical layers, each being insulated from the other by paper, cloth or cooling ducts.

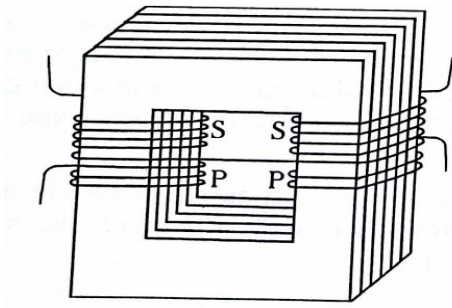


Fig 1: Small transformers

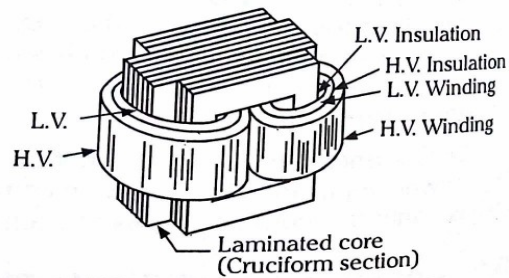
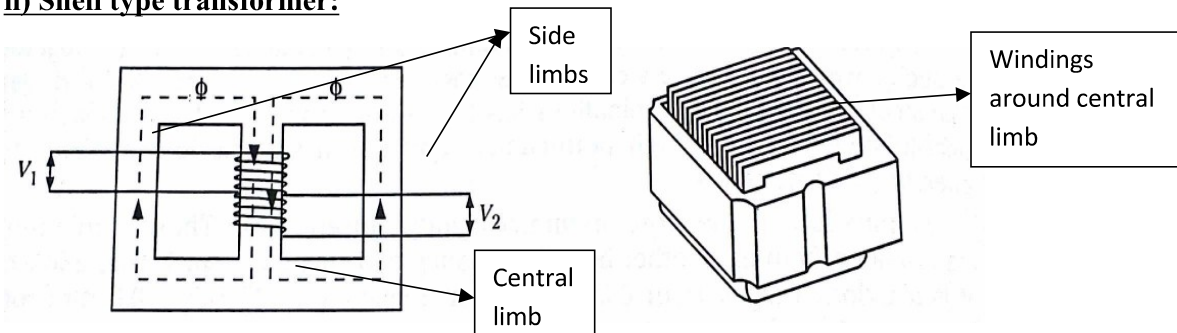


Fig 2: Large transformers

ii) Shell type transformer:

Here, the windings occupy a smaller portion of the core as shown in the figure above. The windings are located on the central limb. The coils are form wound in a disc type insulated from each other by using paper. The insulation spaces between the stacked discs provide horizontal cooling. A commonly used shell type is the rectangular form as shown in figure.

Choice of core or shell type transformers depends on voltage rating, KVA rating of transformer, weight and heat distribution etc. Shell type used for high voltages.

EMF Equation:

Let N_1 = primary turns

N_2 = secondary turns

B_m = maximum flux density in the core (Wb/m²)

A = area of cross section of the wire (m²)

f_1 = frequency of ac input in hertz

Φ = maximum value of flux in the core linking both the windings = $B_m A$

V_1 = applied voltage on primary side

V_2 = secondary terminal voltage

When an alternating voltage, $v_1 = V_m \sin \omega t$ of rms values $V_1 = V_m / \sqrt{2}$ is applied to the primary winding of the transformer, the alternating current flowing through the primary winding produces an alternating flux ϕ , which links with both the primary winding and the secondary winding. Hence, an emf e_1 is induced in the primary winding and emf E_2 is induced in the secondary winding. The equation for $E_1 = -N_1 \frac{d\phi}{dt}$

As the primary voltage is sinusoidal in nature, the current it drives and the resulting flux produced are also sinusoidal. The equation for flux is given by, $\phi = \phi_m \sin \omega t$

Substituting for ϕ in E_1 we get, $E_1 = -N_1 \frac{d\phi}{dt} = -N_1 \omega \phi_m \cos \omega t$

The maximum value of emf induced in the primary winding is given by, $E_{m1} = 2\pi f N_1 \phi_m$

Hence, RMS value, $E_1 = \frac{E_{m1}}{\sqrt{2}} = 4.44 f N_1 \phi_m$

Similarly, $E_2 = 4.44 f N_2 \phi_m$

Dividing equations we get, $\frac{E_1}{E_2} = \frac{N_1}{N_2} = K$

POWER LOSSES IN A TRANSFORMER

The losses occurring in a transformer are

a) core or iron loss and b) copper loss(winding loss)

a) Core or iron losses: these occur due to alternating flux in the transformer core. Because, the core flux remains practically constant for all loads, the core loss is practically constant at all loads. Hence core of transformer is made of material of narrow hysteresis loop.

- i. Hysteresis loss: the power required for magnetization or continuous reversal of flux in the core is dissipated in the form of heat known as hysteresis loss. It depends on supply frequency and flux density.

$$\text{Hysteresis loss} = W_h = P B_{\max}^{1.6} f \text{ watt}$$

where, $B_{\max}$ = maximum flux density(Wb/m²)

f = frequency in Hz

P is a constant

- ii. Eddy current loss: eddy currents occur due to alternating flux and power required to maintain these currents is dissipated in the form of heat known as eddy current loss. These can be minimised by thin laminations in the core, separated by each other from varnish.

$$\text{Eddy current loss} = W_e = Q B_{\max}^2 f^2 \text{ watt},$$

where, Q is a constant

Core losses can be defined as constant losses since flux remains practically constant from no load to full load and also the supply frequency.

Eddy currents: These are the swirling or circular currents set up in a conductor in response to changing magnetic field. This current in turn creates a magnetic field that opposes the cause for it. i.e, it develops a field perpendicular to the changing magnetic field. This tendency of eddy currents to oppose the changing magnetic field causes loss of energy in the form of heat.

b) Copper losses: these losses occur due to the ohmic resistance in both the primary and secondary windings. If R_1 and R_2 are the primary and secondary resistances, and I_1 and I_2 are the primary and secondary currents respectively,

$$\text{Total Cu loss} = I_1^2 R_1 + I_2^2 R_2 = I_1^2 R_{01} + I_2^2 R_{02}$$

where, R_{01} is the equivalent resistance of the transformer referred to primary side. and R_{02} is the equivalent resistance of the transformer referred to secondary side.

EFFICIENCY OF TRANSFORMER:

Efficiency of transformer =

$$= \frac{I_2 V_2 \times \text{p.f.}}{(I_2 V_2 \times \text{p.f.}) + P_C + I_1^2 R_1 + I_2^2 R_2}$$

Or $\eta = 1$

Efficiency is maximum for unity power factor

Condition for maximum efficiency:

$$\text{Copper loss, } W_{cu} = I_1^2 R_{01} + I_2^2 R_{02}$$

where, R_{01} is equivalent resistance of the transformer as referred to primary side

and R_{02} is equivalent resistance of the transformer as referred to secondary side

$$\text{Iron loss} = \text{hysteresis loss} + \text{eddy current loss}$$

$$\text{i.e, } W_i = W_h + W_e$$

Considering primary, primary input = $V_1 I_1 \cos \phi_1$

We know, $\eta =$

$$\eta = \frac{V_1 I_1 \cos \phi_1 - \text{losses}}{V_1 I_1 \cos \phi_1}$$

$$= \frac{V_1 I_1 \cos \phi_1 - I_1^2 R_{01} - W_i}{V_1 I_1 \cos \phi_1}$$

$$\text{or } \eta = 1 - \frac{I_1 R_{01}}{V_1 \cos \phi_1} - \frac{W_i}{V_1 I_1 \cos \phi_1}$$

Differentiating LHS and RHS w.r.t I_1 we have,

$$\frac{d\eta}{dI_1} = 0 - \frac{R_{01}}{V_1 \cos \phi_1} + \frac{W_i}{V_1 I_1^2 \cos \phi_1}$$

For maximum efficiency,

$$\frac{d\eta}{dI_1} \text{ should be equated to zero}$$

$$\text{or } -\frac{R_{01}}{V_1 \cos \phi_1} + \frac{W_i}{V_1 I_1^2 \cos \phi_1} = 0$$

$$\text{or } \frac{W_i}{V_1 I_1^2 \cos \phi_1} = \frac{R_{01}}{V_1 \cos \phi_1}$$

$$\text{or } W_i = I_1^2 R_{01} \quad \text{or} \quad I_2^2 R_{02}$$

$$\text{or } \text{Iron Loss} = \text{Copper Loss}$$

It shows that efficiency of transformer is maximum when iron or constant losses are equal to the copper losses. This is the condition for maximum efficiency.

Examples:

1. A 200 KVA, 10000/400 V, 50 Hz single phase transformer has 100 turns on the secondary. Calculate:

- i) the primary and secondary currents
- ii) the number of primary turns
- iii) the maximum value of flux

solution: full load primary current = 20A

and full load secondary current = 500A

ii) no of primary turns = 2500

iii) $E_2 = 4.44f N_2 \phi_m$

$$400 = 4.44 \times 50 \times 100 \phi_m$$

$$\phi_m = 0.018 \text{ Wb} = 18 \text{ mWb}$$

2. In a 25 KVA, 2000/200 V single phase transformer, the iron and copper losses are 350 watts and 400 watts respectively. Calculate the efficiency at U.P.F at half full load

Solution: power factor, $\cos \phi = 1$

$$\text{Output at half full load} = (12.5 \times \cos \phi) = 12.5 \times 1 = 12.5 \text{ KW}$$

$$\text{Cu loss at half full load} = \left(\frac{1}{2}\right)^2 \times \text{FL loss} = \frac{1}{4} \times 400 = 100 \text{ KW}$$

$$\text{Iron loss} = 350 \text{ W (constant)}$$

$$\text{Total losses} = 100 + 350 = 450 \text{ W} = 0.45 \text{ KW}$$

$$\text{Input} = \text{output} + \text{losses} = 12.5 + 0.45 = 12.95 \text{ KW}$$

$$\text{Efficiency} = \frac{\text{output}}{\text{input}} \times 100 = 96.52\%$$

Three Phase Induction Motors:

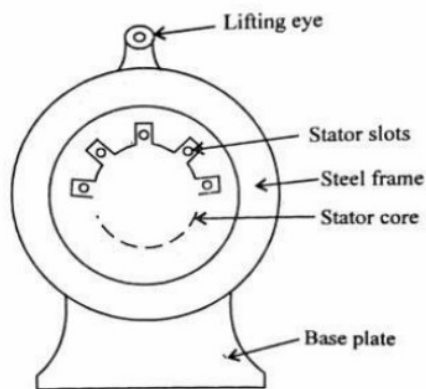
The asynchronous motors or the induction motors are most widely used ac motors in industrial drives. **They convert electrical energy in AC form into mechanical energy.** They work on the principle of electromagnetic induction. The necessary voltage and current in the rotor circuit are produced by induction from the stator, hence called induction motor. They are simple and rugged in construction, quite economical with good operating characteristics and efficiency, requiring minimum maintenance, but have a low starting torque. They run at practically constant speed from no load to full load condition. The 3 - phase induction motors are self-starting while the single phase motors are not self-starting as they produce equal and opposite torques (zero resultant torque) making the rotor stationary. The speed of the squirrel cage induction motor cannot be varied easily.

Construction:

Three phase induction motor consists of two parts. (1) Stator (2) rotor

1. Stator

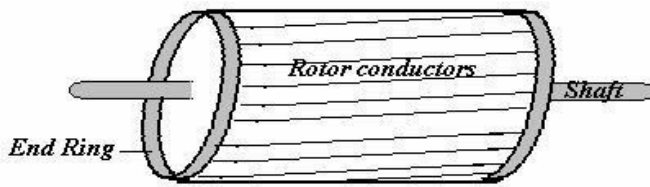
It is the stationary part of the motor supporting the entire motor assembly. This outer frame is made up of a single piece of cast iron in case of small machines. In case of larger machines they are fabricated in sections of steel and bolted together. The core is made of thin laminations of silicon steel and flash enamelled to reduce eddy current and hysteresis losses. Slots are evenly spaced on the inner periphery of the stator core. Conductors are connected to each other to form a balanced 3 - phase star or delta connected stator circuit. They are insulated from each other and placed in stator slots. 3 phase stator is connected to three phase supply



Stator

2. Rotor: It is the rotating part of the motor. Following are the two types of rotor.

i) Squirrel cage rotor: Squirrel cage rotors are widely used because of their ruggedness. The rotor consists of hollow laminated core with parallel slots provided on the outer periphery. The rotor conductors are solid bars of copper, aluminium or their alloys. The bars are inserted from the ends into the semi-enclosed slots and are brazed to the thick short circuited end rings. This sort of construction resembles a squirrel cage hence the name “squirrel cage induction motor”. As the rotor conductors are being permanently short circuited, this prevents the addition of any external

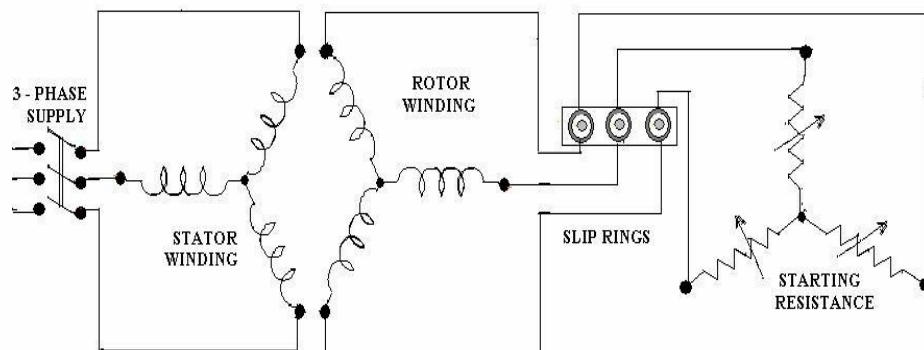


Squirrel cage induction rotor

resistance to the rotor circuit to improve the inherent low starting torque. The rotor bars are not placed parallel to each other but are slightly skewed (inclined at an angle) which reduces the magnetic hum or interference and prevents cogging of the rotor and the stator teeth.

Cogging: Is the condition when stator slots get locked with rotor slots.

ii) Phase wound rotor (Slip ring induction motor):



The rotor in case of a phase wound or slip ring motor has a 3-phase double layer distributed winding made up of coils, similar to that of an alternator. The rotor winding is usually star connected and is wound to the number of stator poles. The terminals are brought out and connected to three slip rings mounted on the rotor shaft with the brushes resting on the slip rings. The brushes are externally connected to the star connected rheostat in case a higher starting torque and modification in the speed torque characteristics are required. Under normal running conditions all the slip rings are automatically short circuited by a metal collar provided on the shaft and the condition is similar to that of a cage rotor. Provision is made to lift the brushes to reduce the frictional losses. The slip ring and the enclosures are made of phosphor bronze.

In both the type of motors the shaft and bearings (ball and roller) are designed for trouble free operation. Fans are provided on the shaft for effective circulation of air. The insulated (mica and

varnish) stator and rotor windings are rigidly braced to withstand the short circuit forces and heavy centrifugal forces respectively. Care is taken to maintain a uniform air gap between the stator and the rotor.

The advantages of squirrel cage rotor are:

1. Rugged in construction and economical.
2. Has a slightly higher efficiency and better power factor than slip ring motor.
3. The absence of slip rings and brushes eliminate the risk of sparking which helps in a totally enclosed fan cooled (TEFC) construction.

The advantages of the slip ring rotor are:

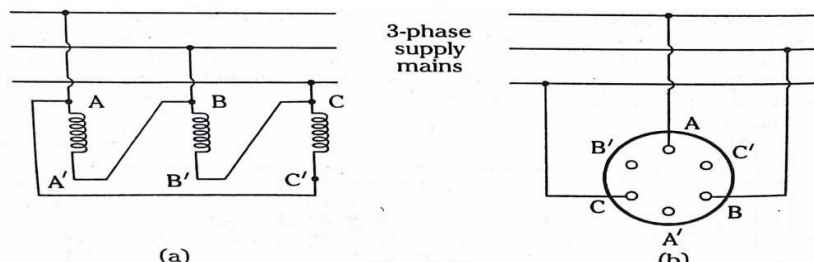
1. The starting torque is much higher and the starting current much lower when compared to a cage motor with the inclusion of external resistance.
2. The speed can be varied by means of solid state switching

Comparison of the squirrel cage and slip ring induction motors:

Squirrel cage IM	Slip ring IM
1.Low starting torque	1.High starting torque
2.Copper loss is less and hence efficiency is more	2.Efficiency is more due to less copper loss
3.Cost of motor is less	3. Cost of motor is high
4.Starting current is high and cannot be reduced.	4.Starting current can be reduced even if high. This is possible by variable resistance connected to rotor winding.
5.Requires less maintenance	5.Maintenance cost is more
6.Used for low starting torque applications. Eg compressors, conveyors, elevators	6.Used for high starting torque drives. Eg lifts, cranes

Rotating magnetic field(RMF): Here we show that the three windings which are 120° electrical apart when supplied with three phase currents, produce magnetic field, that rotates as if poles were rotating.

Let us consider a 3 phase induction motor where delta connected stator windings are supplied with three phase currents to produce a resultant magnetic flux as shown in figure (a). Here, the three phase windings are displaced 120° electrical. The relative positions of the windings AA', BB', CC' is shown in fig (b)



When power is switched on, alternating currents flow in the stator windings which establish its own magnetic flux as shown in fig 1. ϕ_m be the maximum value of the fluxes due to any one phases. The resultant flux ϕ_r at any instant is the vector sum of the individual fluxes, ϕ_R , ϕ_Y and ϕ_B .

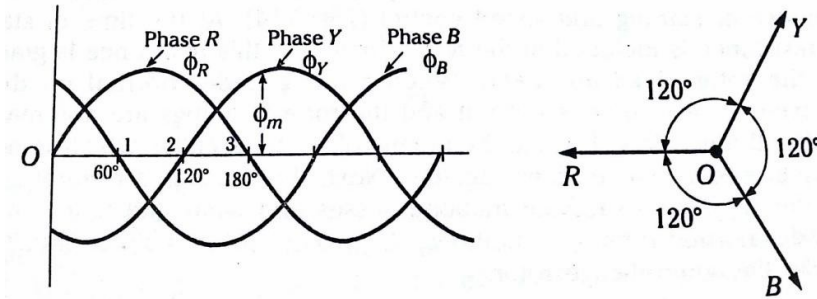


Fig 1

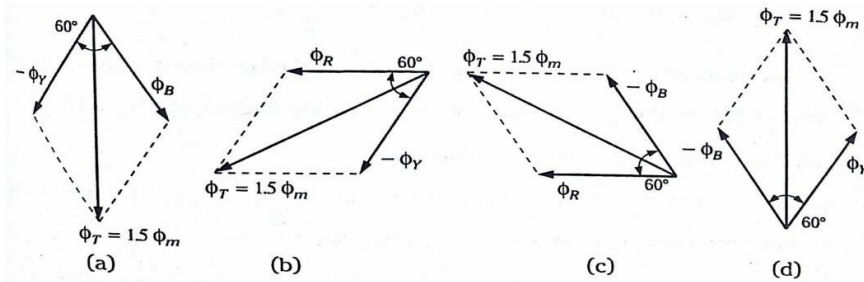


Fig 2

Instantaneous values of three fluxes produced by three phase windings of stator is given by,

$$\phi_R = \phi_m \sin \theta$$

$$\phi_Y = \phi_m \sin(\theta - 120^\circ)$$

$$\phi_B = \phi_m \sin(\theta + 120^\circ)$$

a) At $\theta = 0$ (fig 2a)

This corresponds to point O in the figure 1. Here, $\phi_R = 0$, $\phi_Y = -\phi_m$, $\phi_B = \phi_m$.

Hence, $\phi_r =$

$$= 2 \times \phi_m \cos$$

$$= \times \phi_m$$

$$= \phi_m$$

b) At $\theta = 60^\circ$ (fig 2b). This corresponds to point 1 in the figure 1

Here, $\phi_R = \phi_m$, $\phi_Y = -\phi_m$, $\phi_B = 0$

Therefore, $\phi_r = 2 \times \phi_m \cos = \times \phi_m = \phi_m$

We see that the resultant flux is again ϕ_m , but has rotated clockwise through an angle of 60°

c) At $\Theta = 120^\circ$ (fig 2c). This corresponds to point 2 in the figure 1

Here, $\phi_R = \phi_m, \phi_Y = 0, \phi_B = -\phi_m$

We can again prove that $\phi_T = \phi_m$

We see once again see that the resultant flux is again ϕ_m , but has further rotated clockwise through an angle of 60°

d) At $\Theta = 180^\circ$ (fig 2d). This corresponds to point 3 in the figure 1

Here, $\phi_Y = \phi_m, \phi_R = 0, \phi_B = -\phi_m$

Again $\phi_T = \phi_m$

We see once again see that the resultant flux is again ϕ_m . From figure 2, we see that resultant flux has rotated clockwise through an additional angle of 60° . **Hence, the resultant flux is of constant value ie, 1.5 times the maximum value of the flux due to any phase.** As Θ varies from 0 to 2π , flux also rotates with angular velocity ω and constant magnitude $1.5\phi_m$.

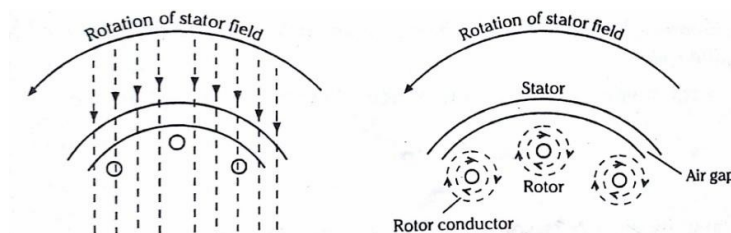
Synchronous speed or stator speed is given by $N_s = \frac{120f}{P}$, where P is the stator poles and f is the supply frequency.

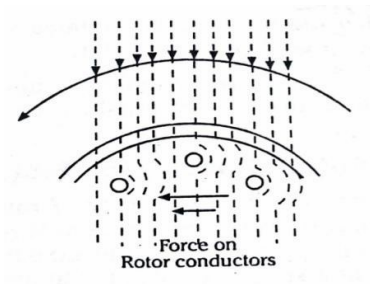
Principle of operation of Induction motor:

When the stator of a 3 phase induction motor is connected to a 3 phase supply, a rotating magnetic field is established which rotates at synchronous speed. The direction of rotation of this field depends on the phase sequence of stator currents and order of connection of the stator terminals to the supply. The constant amplitude flux passes through the air gap and cuts the rotor conductors which are yet stationary.

Due to the relative speed between the rotating flux and stationary rotor conductors, an emf is induced in the latter as per Faraday's law of electromagnetic induction. Frequency of flux is the same as supply frequency, magnitude proportional to the relative velocity between the flux and conductors and direction given by Fleming's Right hand rule. Rotor current is now produced in rotor conductors, which according to Lenz law opposes the cause producing it. Hence to reduce the relative speed, the rotor starts running in the same direction as that of stator.

Generation of torque can be explained as follows: Stator field is rotating in anticlockwise. By Fleming's right hand rule, emf induced in the rotor is outwards. Under both the rotor and stator field, rotor conductors are subjected to force tending them to rotate in anticlockwise direction. Hence rotor rotates in the same direction as stator.





Slip(S):

According to Lenz's law, the direction of rotor current will be such that they tend to oppose the cause producing it. The cause producing the rotor current is the relative speed between the rotating field and the stationary rotor. Hence, to reduce this relative speed, the rotor starts running in the same direction as that of stator field and tries to catch it. In practice the rotor can never reach the speed of the rotating magnetic field produced by the stator. This is because if rotor speed equals the synchronous speed, then there is no relative speed between the rotating magnetic field and the rotor. This makes the rotor current zero and hence no torque is produced and the rotor will tend to remain stationary. In practice, windage and friction losses cause the rotor to slow down. Hence, the rotor speed (N) is always less than the stator field speed (N_s). Thus the induction motor cannot run with ZERO SLIP.

The difference between the synchronous speed (N_s) of the rotating stator field and the actual rotor speed (N) is called the slip.

Slip speed = $(N_s - N)$ r.p.m

% Slip (s) = $\frac{N_s - N}{N_s} \times 100$

$N_s - N$ is sometimes called the slip speed. Hence rotor or motor speed is $N = N_s(1 - s)$

No load to full load slip is 3-6% only. Hence, Induction motor is a constant speed motor.

SIGNIFICANCE OF SLIP IN AN INDUCTION MOTOR.

OR

CAN $N=N_s$?

When rotor starts rotating, it tries to catch the speed of rotating magnetic field. If it catches the speed of rotating magnetic field, the relative motion between rotor and the rotating magnetic field will vanish ($N_s - N = 0$)

But due to inertia of rotor, this does not happen in practice and rotor continues to rotate with a speed slightly less than the synchronous speed of the rotating magnetic field in the steady state.

The induction motor never rotates at synchronous speed. The speed at which it rotates is hence called sub synchronous speed and motor sometimes called asynchronous motor.

$$N < N_s$$

So it can be said that the rotor slips behind the rotating magnetic field produced by stator. The difference between the two is called slip speed of the motor.

$$N_s - N = \text{slip speed of the motor in rpm}$$

Frequency of the rotor current:

At standstill, the frequency of the rotor current or emf is the same as the supply frequency. When motor starts running, the frequency of the induced emf and hence the rotor current varies with the slip and is proportional to it. Let at any speed N of the rotor, the frequency of the rotor current be f' .

Then $N_s - N = \dots$ also $N_s = \dots$

Dividing we get, $\dots = S$

Therefore, $f' = S \cdot f$

i.e, frequency of rotor current is slip times the supply frequency.

Squirrel cage induction motor:

Squirrel cage induction motors are simple and rugged in construction, are relatively cheap and require little maintenance. Hence, squirrel cage induction motors are preferred in most of the industrial applications such as in

- i) Lathes
- ii) Drilling machines
- iii) Agricultural and industrial pumps
- iv) Industrial drives.

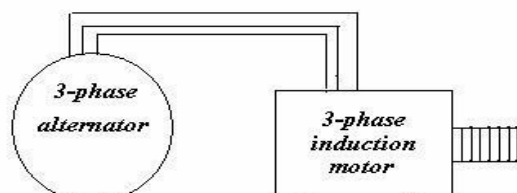
Slip ring induction motors:

Slip ring induction motors when compared to squirrel cage motors have high starting torque, smooth acceleration under heavy loads, adjustable speed and good running characteristics. They are used in

- i) Lifts
- ii) Cranes
- iii) Conveyors, etc.,

Example:

1. A 6 pole induction motor is supplied by a 10 pole alternator, which is driven at 600 rpm. If the induction motor is running at 970 rpm, determine its percentage slip.



Solution: From alternator data:

$$F = 50 \text{ Hz}$$

Synchronous speed of the motor, $N_s = 1000 \text{ rpm}$

$$\% \text{ slip} = \frac{N_s - N}{N_s} \times 100 = 3\%$$